MATS

Failure Modes when Training with Distribution Shift

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MOTIVATION

Safety post-training — alignment can be shallow if the distribution shift between pretraining and post-training impairs what is learned.

Deployment-time revisions — continuously fixing model failures observed during deployment requires responsiveness to distribution shifts.

Interpretability — Investigating this phenomena can generally improve our understanding of model internals and generalization.

PRIOR WORK

Warm-start gap [1] — expanding the training data incrementally hurts performance vs. training on the full dataset from scratch.

Overtraining language models [4] — extended pretraining past a threshold can hurt performance after instruction finetuning.

Critical learning periods [2,3] — if learning during a critical early period is hindered, the model never catches up.

Loss of plasticity [5] — under a continuously changing data distribution, models lose the ability to adapt.

We investigate the causes of these phenomena with a Bayesian perspective.

ORDERING EFFECTS

Non-pathological ordering effects

A neural network can be less adaptable to a distribution shift purely because the weights concentrate on the initial distribution and then have to move farther for the second distribution.

Setting: a teacher-student deep linear network with student $W_D W_{D-1} \dots W_1$ and teachers

$$T_A = \mathrm{diag}(4,3,2,1,0,0,0,0), \qquad T_B = \mathrm{diag}(0,0,0,0,4,3,2,1).$$

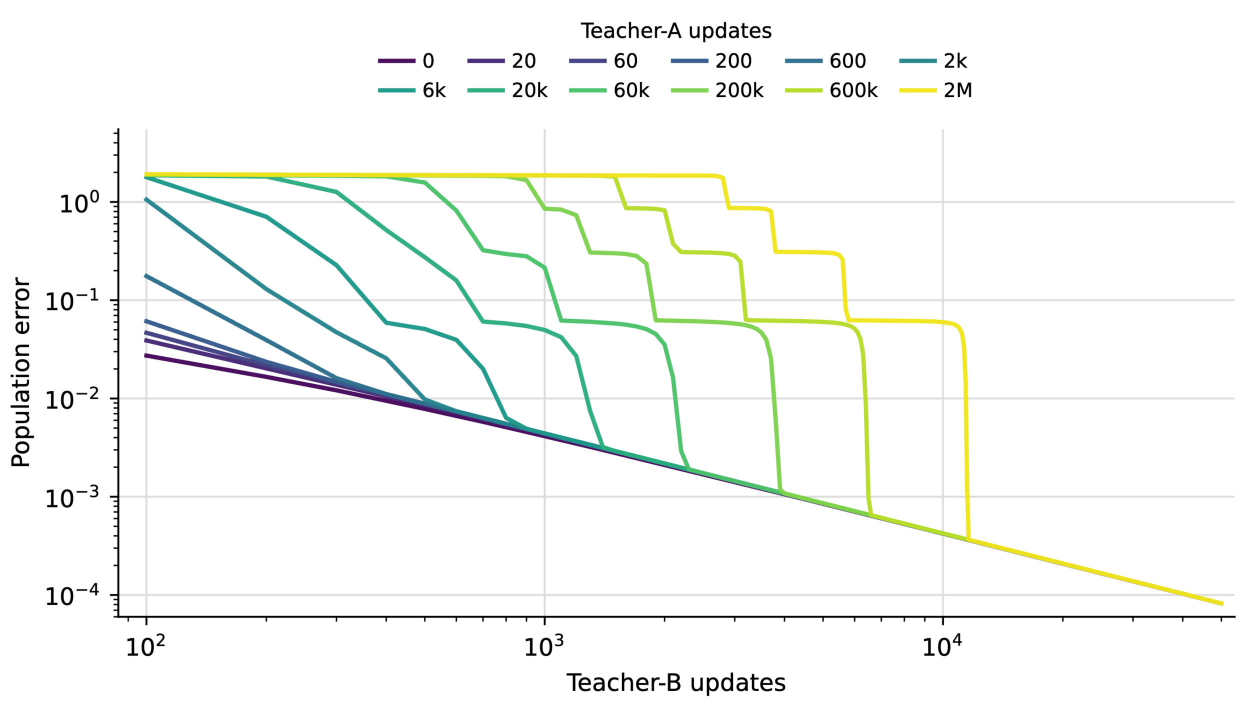

Figure 1. Population error curves when training a student network on T_B, initialized as the weights after training on a specified number of training steps on T_A.

Pathological ordering effects

Setting: a transformer performs in-context linear regression, with data generated as

$$y = w^T x + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \qquad w \sim \mathrm{Unif}(T), \quad T = \{w_0, \dots, w_k\}.$$

The model predicts

$$y_i \mid x_i, (x_0, y_0, \dots, x_{i-1}, y_{i-1}).$$

In the latter stages of training on the initial task distribution T_A, the model undergoes layernorm collapse. Unlike the non-pathological case, this loss of adaptability may not be recoverable.

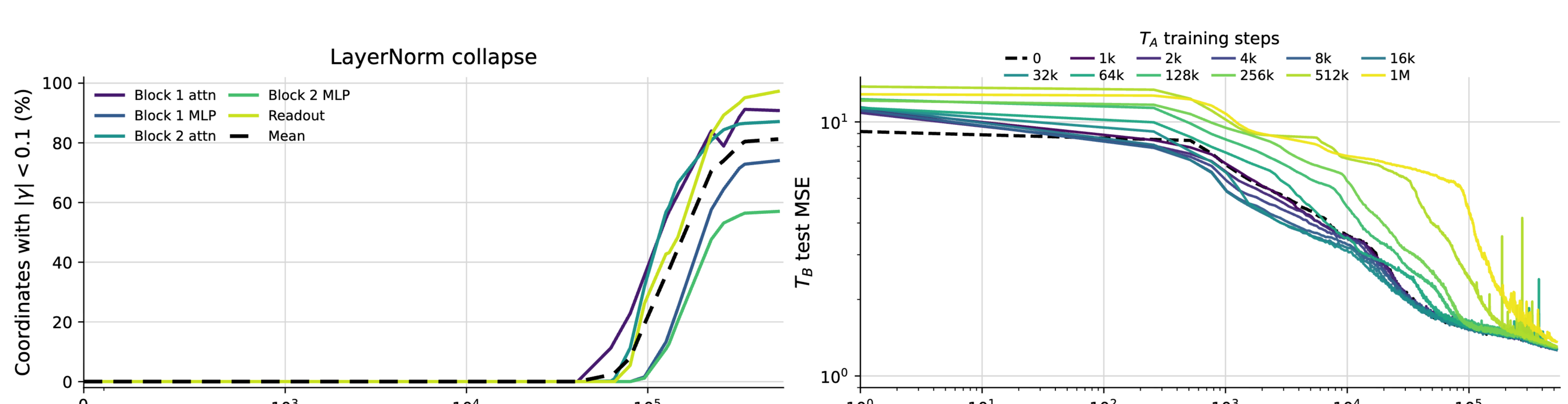

Figure 2. *Left:* Percentage of layer normalization coordinates below threshold 0.1, during training on T_A. *Right:* Test MSE curves when training on a secondary task distribution T_B, initialized as the weights after training on a specified number of training steps on T_A.

DATA EFFECTS

Bayesian learners are invariant to data order, but their performance can still be hurt by the extraneous data from before a distribution shift.

Model: a 2 layer linear network $\mathbb{R}^{2d} \mapsto \mathbb{R}^d$.

$$\hat{y} = W_2 W_1 x = W_2 \begin{pmatrix} W_1' \\ W_1'' \end{pmatrix} \begin{pmatrix} x' \\ x'' \end{pmatrix}, \qquad \Pr(y_i \mid x_i, w) = \exp(-(y_i - W_2 W_1 x_i)^2)$$

The primary dataset $D'_{n'}$, where the ground truth is low rank, $\mathrm{rank}(W'_\star) = \frac{d}{2}$:

$$x_i = \begin{pmatrix} x_i' \\ 0 \end{pmatrix} \quad x_i' \sim \mathcal{N}(0, I_d) \quad y_i = W'_\star x_i' + \xi_i \quad \xi_i \sim \mathcal{N}(0, \sigma^2 I_d)$$

The secondary dataset $D''_{n''}$, where the ground truth is full rank, $\mathrm{rank}(W''_\star) = d$:

$$x_i = \begin{pmatrix} 0 \\ x_i'' \end{pmatrix} \quad x_i'' \sim \mathcal{N}(0, I_d) \quad y_i = W''_\star x_i'' + \xi_i \quad \xi_i \sim \mathcal{N}(0, \sigma^2 I_d)$$

The Bayes generalization error, where x, y are sampled from the true distribution that generates the primary dataset, and $\Pr(w \mid D'_{n'}, D''_{n''})$ is the Bayesian posterior after updating on n' datapoints from $D'_{n'}$ and n'' datapoints from $D''_{n''}$:

$$G(n', n'') = \mathbb{E}_{x,y}\, \mathbb{E}_{D'_{n'}, D''_{n''}} \left[\int \Pr(y \mid x, w) \Pr(w \mid D'_{n'}, D''_{n''}) dw \right]$$

Theorem *The generalization error when only training on n datapoints from the primary distribution is*

$$G(n, 0) = \frac{15}{16} \cdot \frac{d^2}{2} \cdot \frac{1}{n} + O(n^{-2}).$$

Meanwhile, the error when training on both n primary datapoints and a proportional amount, αn, of secondary data is

$$G(n, \alpha n) = \frac{d^2}{2} \cdot \frac{1}{n} + O(n^{-2}).$$

When the secondary data is added, the speed of convergence slows down by a constant factor.

Intuition. Even though the model is capable of optimal performance simultaneously on both datasets, the secondary dataset $D''_{n''}$ causes the learner to converge to a different global minimum that's surrounded by a less flat loss landscape, and so the learner suffers more from finite-sample noise and thus converges slower.

COMPARING DATA & ORDERING EFFECTS

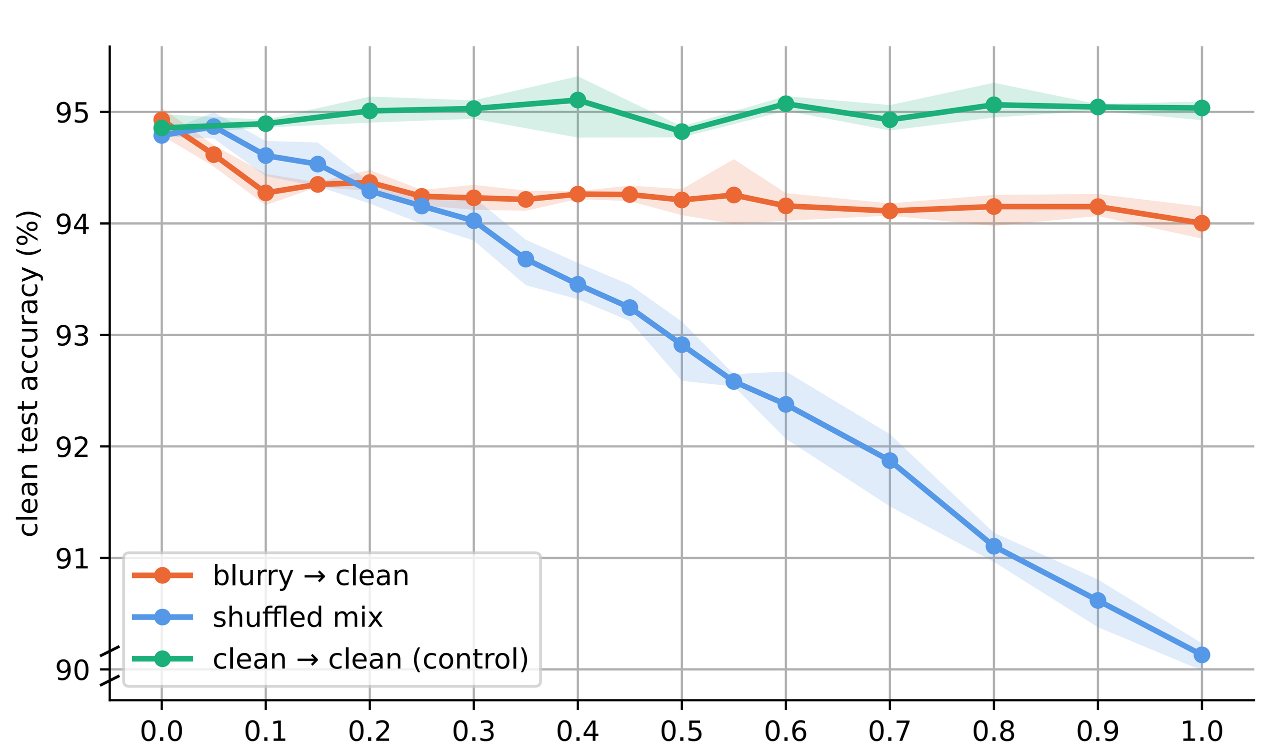

Figure 3. Training on $p \cdot 400$ blurry then 400 clean epochs on CIFAR10 (orange) is worse than clean-only training (green), but better than shuffling the same proportion of blurry and clean data (blue).

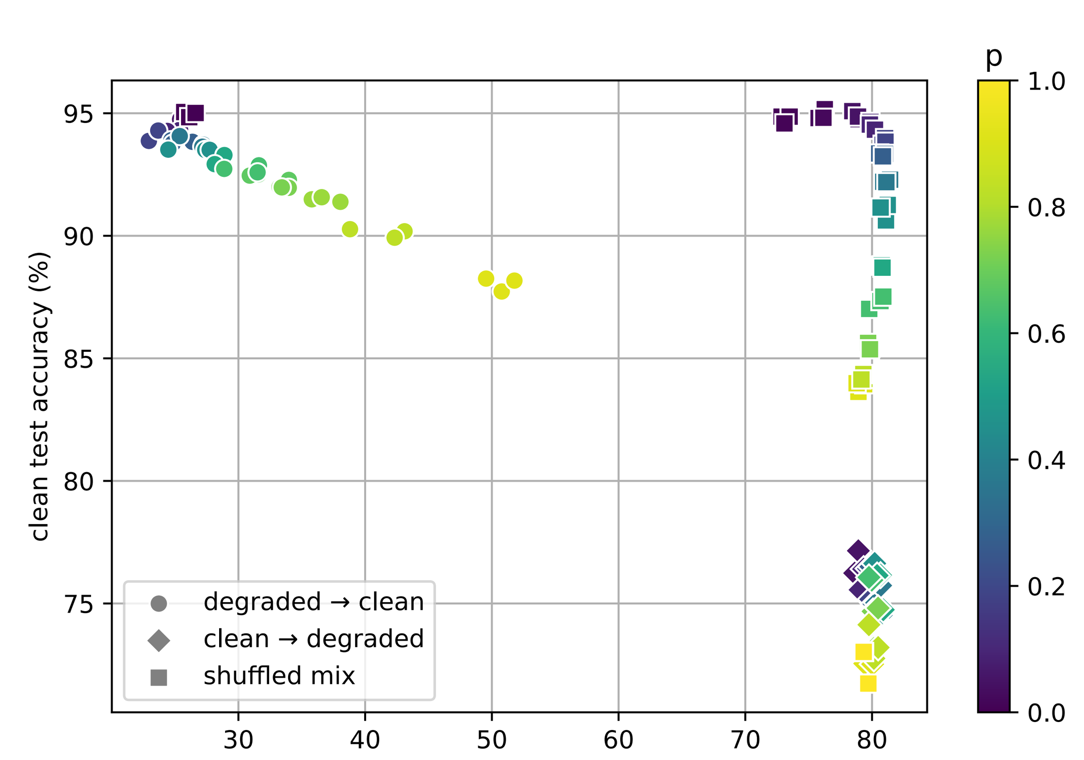

Figure 4. Performance on clean and blurry CIFAR10 after 400 epochs training on $(1-p) \cdot 400$ clean images and $p \cdot 400$ blurry images. Models trained on blurry first and then clean are far from pareto optimal.

NEXT STEPS

1. Decouple between pathological and non-pathological ordering effects in loss of plasticity.
2. Test Bayesian explanations of loss of plasticity (Type-B phase transitions in SLT).

[1] Ash & Adams. On Warm-Starting Neural Network Training. NeurIPS 2020.
[2] Kleinman, Achille & Soatto. Critical Learning Periods Emerge Even in Deep Linear Networks. ICLR 2024.
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[4] Springer et al. Overtrained Language Models Are Harder to Fine-Tune. ICML 2025.
[5] Dohare et al. Loss of Plasticity in Deep Continual Learning. Nature 2024.
[6] Pepin Lehalleur & Rimányi. Geometry of Fibers of the Multiplication Map of Deep Linear Neural Networks. arXiv:2411.19920, 2024.